RESEARCH PAPER: RETRIEVAL AUGMENTED GENERATION SYSTEMS

1. Abstract and Foundations

Retrieval-Augmented Generation (RAG) is an architectural approach that enhances question-answering performance by grounding responses in external knowledge sources. By indexing document passages and retrieving relevant context dynamically, RAG prevents unsupported hallucinations and ensures factual accuracy based on verified records.
In this system, TF-IDF vectorization and cosine similarity are employed as a deterministic, fully local baseline for passage retrieval.

2. Vector Representations and Term Weighting

Term Frequency-Inverse Document Frequency (TF-IDF) evaluates the relative importance of words within a corpus. Sublinear term frequency scaling helps moderate the impact of overly repetitive terms. When coupled with stop-word removal and n-gram representations, TF-IDF efficiently matches lexical query patterns to document segments.

3. Evaluation and Hallucination Prevention

A primary requirement of trustworthy question answering is knowing when not to answer.
When a query lacks supporting evidence in the document corpus, the retrieval confidence
falls below the designated cosine similarity threshold. Under this condition, the system
returns the exact failure indicator ANSWER_NOT_FOUND_IN_DOCUMENT rather than speculating.

4. Conclusion and Future Directions

Deterministic extractive baselines offer complete explainability, high inference speeds,
and zero reliance on external cloud APIs or proprietary model weights.